

16. (1.5 points)

Given the following financial information for an insurance company as of December 31, 2018:

- Beginning US GAAP equity as of January 1, 2019 = \$195 million.
- US GAAP net income forecast for 2019 = \$43 million.
- Change in reserves included in net income forecast for 2019 = \$11 million.
- Capital needed to maintain target rating at December 31, 2019 = \$220 million.
- Risk free interest rate = 3.0%.
- Expected equity market risk premium = 6.5%.
- Insurance company equity beta = 0.8.

a. (0.5 point)

Calculate the Free Cash Flow to Equity (FCFE) for 2019.

b. (0.5 point)

Calculate the reinvestment rate for 2019.

c. (0.5 point)

Briefly describe two reasons why the FCFE approach is preferred to the Free Cash Flow to the Firm (FCFF) approach for valuing insurance companies.

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SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 16	
TOTAL POINT VALUE: 1.5	LEARNING OBJECTIVE(S): B1, B2
SAMPLE ANSWERS	
Part a: 0.5 point	
<u>Sample 1</u> $FCFE = 43 - (220 - 195) = \$18M$	
<u>Sample 2</u> $k = .03 + .8 * .065 = .082$ $18 / 1.082 = \$16.64M$	
Part b: 0.5 point	
<u>Sample 1</u> $(220 - 195) / 43 = 58.14\%$	
<u>Sample 2</u> $g = ROE * p$ $g = (220 - 195) / 195; ROE = 43 / 195; p = 25 / 43 = 58.14\%$	
Part c: 0.5 point	
<u>Sample 1</u> - FCFF values equity and debt combined and then subtracts the market value of the debt to get to the equity. This is a complication for insurance companies because policyholder liabilities are indistinguishable from other forms of debt. - Policyholder liabilities make it difficult to calculate the WACC or APV for discounting.	
<u>Sample 2</u> Both reasons have to do with the difficulty in distinguishing between debt/policyholder liability. - WACC needs an estimate of debt - The FCFF approach subtracts market value of debt	
EXAMINER'S REPORT	
Candidates were expected to understand how to use the Discounted Cash Flow (DCF) Valuation using free cash flow to equity (FCFE), including reasons why this method is preferred over the free cash flow to the firm (FCFF) method for P&C insurers.	
Part a	
Candidates were expected to correctly calculate the FCFE for 2019. Credit was given if the candidate had the correct answer but added an additional step in discounting the FCFE. Common mistakes included: - Only calculating change in required surplus - Incurred adjustments for reserve changes as no adjustment should be made	
Part b	
Candidates were expected to correctly calculate the reinvestment rate for 2019. Common mistakes included: - Calculating the growth rate - Calculating the required return on equity	

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Part c
Candidates were expected to understand that policyholder liabilities can be viewed as arbitrary from debt and therefore make the FCFF difficult to calculate due to the necessary subtraction of debt in the formula as well as the required discount rate of the debt.